Complete Guide to Frame Dominance Measurement

A Comprehensive, Pedagogical Explanation of the Four-Component Method

Conference Presentation Reference Document

Detailed Technical and Theoretical Documentation

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1 Introduction: Why We Need This Framework

1.1 The Fundamental Problem

Imagine you're analyzing 47 years of newspaper articles about climate change - over 260,000 articles, 9.2 million sentences. You notice that journalists use different "frames" or interpretive lenses:

- Sometimes they discuss climate as a **scientific** issue ("Scientists report rising temperatures")
- Sometimes as a **political** issue ("Parliament debates carbon tax")
- Sometimes as an **economic** issue ("Green energy creates jobs")
- And so on...

The critical question is: **Which frames are truly dominant?** Not just frequent, but frames that fundamentally structure how climate change is discussed.

1.2 What Makes a Frame "Dominant"?

Think of frames like instruments in an orchestra. Some instruments (frames) play more often (frequency), but that doesn't make them dominant. The conductor (dominant frame) might not make the most sound, but they organize everything else. We need to identify the "conductors" of climate discourse.

A dominant frame should:

1. **Carry information**: When it appears, it tells us a lot about what else will appear
2. **Occupy central positions**: It connects to and influences other frames
3. **Drive temporal dynamics**: It causes other frames to appear
4. **Maintain consistent presence**: It's reliably there, not just episodic

1.3 Our Solution: Four Complementary Lenses

We examine each frame through four different analytical lenses, each capturing a different aspect of dominance. Think of it like evaluating a student: we don't just look at test scores, but also participation, leadership, and consistency. Only frames that excel in at least 3 out of 4 dimensions are considered dominant.

2 Component 1: Information Theory Analysis

2.1 The Big Idea

Information theory, developed by Claude Shannon in 1948, quantifies information and uncertainty. In our context, we ask: "How much does knowing about one frame reduce our uncertainty about others?"

2.2 Understanding Entropy: The Foundation

2.2.1 What is Entropy?

Entropy measures uncertainty or "surprise" in a system.

Intuitive Example

Imagine a coin flip:

- Fair coin: Maximum entropy (1 bit) - you're maximally uncertain
- Two-headed coin: Zero entropy (0 bits) - no uncertainty at all
- Biased coin (70% heads): Medium entropy (0.88 bits)

Mathematically:

$$H(X) = -\sum_i p(x_i) \log_2 p(x_i)$$

For frames, entropy tells us how predictable the discourse is. High entropy = many different frame combinations. Low entropy = predictable patterns.

2.3 Metric 1: Mutual Information (MI)

2.3.1 What It Measures

Mutual information quantifies how much knowing one frame tells us about another. It's the "shared information" between frames.

2.3.2 The Mathematics

The formula:

$$MI(X;Y) = \sum_{x,y} p(x,y) \log_2 \left(\frac{p(x,y)}{p(x)p(y)} \right)$$

Let's break this down:

- $p(x,y)$ = probability of seeing frames X and Y together
- $p(x)p(y)$ = probability if they were independent
- The ratio tells us how much more likely they are to co-occur than by chance

2.3.3 Concrete Example

Let's say we're analyzing a week of coverage:

- Political frame appears in 40% of sentences
- Economic frame appears in 30% of sentences
- They appear together in 25% of sentences

If independent: $0.4 \times 0.3 = 0.12$ (12% expected co-occurrence)
Actually observed: 0.25 (25% actual co-occurrence)
They co-occur MORE than expected $\rightarrow$ positive mutual information

2.3.4 Implementation Details

Technical Implementation

Step 1: Discretization
We convert continuous proportions to discrete bins:

- Number of bins: $n_{bins} = \min(10, \max(3, \lfloor\sqrt{N}\rfloor))$
- This adaptive binning ensures appropriate granularity
- Example: 100 weeks $\rightarrow$ 10 bins; 9 weeks $\rightarrow$ 3 bins

Step 2: Compute Pairwise MI
For each frame pair (i,j):

- Build contingency table of co-occurrences
- Calculate MI using the formula above
- Store in symmetric matrix

Step 3: Average Across All Pairs
Each frame's MI score = average MI with all other frames

2.3.5 Interpretation for Conference

High MI means the frame is an "information hub":

- Political frame has high MI $\rightarrow$ knowing it's present tells us a lot about other frames
- It's like a keystone species in ecology - its presence/absence changes everything

2.4 Metric 2: Entropy Reduction

2.4.1 The Core Question

"How much does knowing this frame's state reduce uncertainty about the entire system?"

2.4.2 Step-by-Step Calculation

Detailed Entropy Reduction Calculation

Step 1: Calculate Joint Entropy H(X)
This is the entropy of ALL frames together:

$$H(X) = -\sum p(x_1, x_2, ..., x_8) \log_2 p(x_1, x_2, ..., x_8)$$

Example: With 8 frames, there are $2^8 = 256$ possible configurations.
We calculate the probability of each configuration and compute entropy.
Step 2: Calculate Conditional Entropy H(X—Y)
For a specific frame Y:

$$H(X|Y) = \sum_y p(y) H(X|Y = y)$$

This means:

- Calculate entropy when Y is present: $H(X|Y = 1)$
- Calculate entropy when Y is absent: $H(X|Y = 0)$
- Weight by probabilities: $p(Y = 1) \cdot H(X|Y = 1) + p(Y = 0) \cdot H(X|Y = 0)$

Step 3: Compute Reduction

$$ER = \frac{H(X) - H(X|Y)}{H(X)}$$

This gives us a percentage: "Knowing Y reduces uncertainty by ER%"

2.4.3 Worked Example

Imagine a simplified 3-frame system:

Without any information:

- All 8 configurations possible (2^3)
- High entropy: $H(X) = 2.5$ bits

Knowing Political frame is present:

- Economic frame: 80% likely present
- Scientific frame: 60% likely present
- Entropy drops to: $H(X|Political = 1) = 1.2$ bits

Knowing Political frame is absent:

- Economic frame: 20% likely present
- Scientific frame: 70% likely present
- Entropy: $H(X|Political = 0) = 1.8$ bits

Final calculation:

- P(Political=1) = 0.6, P(Political=0) = 0.4
- $H(X|Political) = 0.6 \times 1.2 + 0.4 \times 1.8 = 1.44$ bits
- Reduction $= (2.5 - 1.44)/2.5 = 0.424 = 42.4\%$

The political frame reduces system uncertainty by 42.4%!

2.4.4 Why This Matters

Frames with high entropy reduction are "organizing principles" of discourse:

- They create predictable patterns
- Other frames align around them
- They structure the discourse space

2.5 Metric 3: Information Content

2.5.1 What It Measures

The self-entropy of a frame's distribution over time. High variability = high information content.

2.5.2 The Formula

$$H(Frame_i) = -\sum_t p(frame_i, t) \log_2 p(frame_i, t)$$

2.5.3 Interpretation

- A frame that's always 50% present: Low information (boring, predictable)
- A frame that varies 0%-100%: High information (dynamic, informative)

This balances against frames that are simply "always there" without adding structure.

3 Component 2: Network Analysis

3.1 Conceptual Foundation

We treat discourse as a network:

- **Nodes** = Frames
- **Edges** = Co-occurrence relationships
- **Edge weights** = Strength of co-occurrence

3.2 Building the Network

3.2.1 Temporal Windows

Network Construction Process

1. **Create Rolling Windows**
   - Window size: 52 weeks (1 year)
   - Overlap: 50% (26 weeks)
   - Result: Smooth evolution tracking
2. **Determine Edges** For each window, create edge between frames i and j if:
   - Both exceed 0.1 proportion threshold in same week
   - Edge weight $= P(i > 0.1 \cap j > 0.1)$
3. **Filter Noise**
   - Remove edges below significance threshold
   - Keep only meaningful co-occurrences

3.2.2 Example Network

Week 1: Political (35%), Economic (28%), Scientific (15%)

- Political-Economic edge: Strong (both ¿ 10%)
- Political-Scientific edge: Strong (both ¿ 10%)
- Economic-Scientific edge: Strong (both ¿ 10%)

Over 52 weeks, we count how often each pair co-occurs above threshold.

3.3 Metric 1: PageRank Centrality

3.3.1 The Concept

Google's PageRank algorithm applied to frames. A frame is important if it's connected to other important frames.

3.3.2 The Algorithm

PageRank Step-by-Step

Initialization: Each frame starts with equal PageRank = 1/N
Iteration:

$$PR(i) = \frac{1-d}{N} + d \sum_{j \in M(i)} \frac{PR(j)}{L(j)}$$

Where:

- $d = 0.85$ (damping factor - probability of following links)
- N = number of frames (8)
- $M(i)$ = frames linking to frame i
- $L(j)$ = number of outbound links from frame j

Convergence: Iterate until values stabilize (typically 20-30 iterations)

3.3.3 Intuitive Example

Imagine frames as websites:

- Political frame = Wikipedia (everyone links to it)
- Economic frame = Major news site (many important links)
- Minor frame = Personal blog (few links)

PageRank identifies the "Wikipedias" of climate discourse.

3.3.4 Why Political Frame Scores High

The political frame has high PageRank because:

1. It co-occurs with Economic (high PageRank)
2. It co-occurs with Scientific (medium PageRank)
3. These important frames "vote" for Political
4. This creates a reinforcing cycle

3.4 Metric 2: Eigenvector Centrality

3.4.1 The Difference from PageRank

While PageRank considers link structure, eigenvector centrality considers connection strength.

3.4.2 Mathematical Foundation

Eigenvector centrality is the principal eigenvector of the adjacency matrix:

$$Ax = \lambda x$$

Where:

- A = adjacency matrix (connection strengths)
- x = centrality scores
- λ = largest eigenvalue

3.4.3 Interpretation

"It's not just who you know, but how strongly you're connected to important nodes"

- PageRank: Binary connections (yes/no)
- Eigenvector: Weighted connections (how strong)

3.5 Metric 3: Temporal Stability

3.5.1 The Calculation

$$Stability = \frac{1}{CV} = \frac{\mu}{\sigma}$$

Where CV is coefficient of variation across time windows.

3.5.2 Why It Matters

A frame might be central during one period but disappear later:

- Environmental frame: High centrality during disasters, low otherwise
- Political frame: Consistently central across decades

Stability rewards consistent centrality, not temporary spikes.

4 Component 3: Causality Analysis

4.1 The Fundamental Question

"Which frames drive the appearance of others?" We want to identify frames that don't just correlate but actually cause other frames to appear.

4.2 Method 1: Granger Causality

4.2.1 The Concept

Granger causality tests whether past values of X help predict future values of Y, beyond what Y's own past provides.

4.2.2 The Statistical Test

Granger Causality Procedure

Step 1: Baseline Model Predict Y using only its past:

$$Y_t = \alpha + \sum_{i=1}^{p} \beta_i Y_{t-i} + \epsilon_t$$

Calculate residual sum of squares: RSS_1
Step 2: Full Model Predict Y using its past AND X's past:

$$Y_t = \alpha + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{j=1}^{p} \gamma_j X_{t-j} + \epsilon_t$$

Calculate residual sum of squares: RSS_2
Step 3: F-Test

$$F = \frac{(RSS_1 - RSS_2)/p}{RSS_2/(n - 2p - 1)}$$

If F is significant, X Granger-causes Y

4.2.3 Adaptive Lag Selection

We adapt lag order to data availability:

Data Points	**Maximum Lag**
N ¡ 10	1 lag
10 $\leq$ N ¡ 20	2 lags
20 $\leq$ N ¡ 40	3 lags
N $\geq$ 40	4 lags

Within maximum, AIC selects optimal lag.

4.2.4 Example: Political $\rightarrow$ Economic

Week 1: Political = 0.35
Week 2: Political = 0.40, Economic = 0.25
Week 3: Political = 0.38, Economic = 0.30
Week 4: Political = 0.32, Economic = 0.28
Does Political at t-1 help predict Economic at t?

- Baseline: Economic predicted from its own past
- Enhanced: Economic predicted from its past + Political's past
- If enhanced model significantly better $\rightarrow$ Granger causality

4.2.5 Interpretation of Results

For each frame, we calculate:

- **Breadth**: What percentage of other frames does it Granger-cause?
- **Strength**: Average F-statistic for significant relationships

Political frame shows:

- Breadth: 75% (causes 6 out of 8 other frames)
- Strength: High F-statistics (strong predictive power)

4.3 Method 2: Transfer Entropy

4.3.1 Beyond Linear Relationships

Granger assumes linear relationships. Transfer entropy captures any relationship, linear or not.

4.3.2 The Mathematics

Transfer Entropy Formula

$$T_{X\to Y} = \sum p(y_{t+1}, y_t^k, x_t^l) \log \frac{p(y_{t+1}|y_t^k, x_t^l)}{p(y_{t+1}|y_t^k)}$$

Breaking this down:

- y_t^k = past k values of Y
- x_t^l = past l values of X
- Numerator: Probability of Y's future given both histories
- Denominator: Probability of Y's future given only Y's history

The ratio measures information gain from knowing X's history.

4.3.3 Intuitive Understanding

Think of it as "information flow":

- If X's past helps predict Y's future → information flows X → Y
- Measured in bits (like mutual information)
- Directional: $T_{X\to Y} \neq T_{Y\to X}$

4.3.4 Example Calculation

Simplified 2-state system (present/absent):

Data:

- When Political was present yesterday, Economic is present today 70% of time

- When Political was absent yesterday, Economic is present today 30% of time
- Economic's own persistence: 60% (if present yesterday, likely present today)

Transfer Entropy:

- Knowing Political's past changes Economic's predictability
- From 60% baseline to 70% or 30% depending on Political
- This difference in predictability = transfer entropy

4.3.5 Why Use Both Methods?

Aspect	**Granger**	**Transfer Entropy**
Relationships	Linear only	Any type
Statistical power	Higher	Lower
Interpretation	Regression coefficients	Information bits
Computation	Fast	Slower

Together they provide comprehensive causality picture.

5 Component 4: Proportional Dominance

5.1 The Rationale

Frequency matters, but it's not everything. We need to distinguish:

- Frames that are consistently present vs. episodic
- Frames above baseline vs. average
- Stable presence vs. volatile

5.2 Metric 1: Mean Proportion

5.2.1 Calculation

$$\bar{p}_i = \frac{1}{T}\sum_{t=1}^{T} p_{i,t}$$

Simply the average proportion across all time periods.

5.2.2 Example Values

- Political: 35% average presence
- Economic: 28% average presence
- Scientific: 22% average presence
- Environmental: 8% average presence

5.3 Metric 2: Elevation Score

5.3.1 The Concept

How much does a frame exceed the median of all frames?

5.3.2 Formula

$$Elevation = \max\left(0, \frac{p_i - median}{median}\right)$$

5.3.3 Example Calculation

- Median proportion across all frames: 15%
- Political frame average: 35%
- Elevation $= (35 - 15)/15 = 1.33$ (133% above median)

5.3.4 Why Not Just Use Mean?

Elevation normalizes by median, making it robust to outliers and comparable across different datasets.

5.4 Metric 3: Consistency Score

5.4.1 Formula

$$Consistency = \frac{1}{1 + CV} = \frac{1}{1 + \sigma/\mu}$$

5.4.2 Interpretation

High consistency means reliable presence:

- Political frame: $\mu = 0.35$, $\sigma = 0.10$, CV $= 0.29$, Consistency $= 0.78$
- Environmental frame: $\mu = 0.08$, $\sigma = 0.08$, CV $= 1.0$, Consistency $= 0.50$

Political is more consistent despite both having same standard deviation.

5.5 Metric 4: Percentile Rank

5.5.1 Calculation

Where does frame rank in distribution of all mean proportions?

5.5.2 Example

8 frames ranked by mean proportion:

1. Political: 35% $\rightarrow$ 100th percentile
2. Economic: 28% $\rightarrow$ 87.5th percentile
3. Scientific: 22% $\rightarrow$ 75th percentile

4. ...
5. Security: 3% $\rightarrow$ 12.5th percentile

6 Integration: Bringing It All Together

6.1 The Composite Score

6.1.1 Step 1: Normalize Components

Each component produces raw scores. We normalize to [0,1]:

Normalization Procedure

For each component:

1. Rank frames by component score
2. Apply min-max normalization: $\frac{score-min}{max-min}$
3. Result: Highest = 1.0, Lowest = 0.0

6.1.2 Step 2: Equal Weighting

$$Composite = 0.25 \times Info + 0.25 \times Network + 0.25 \times Causality + 0.25 \times Proportional$$

Why equal weights?

- No theoretical reason to privilege one dimension
- Robustness check shows results stable to weight variations
- Transparency and simplicity for replication

6.2 The Consensus Approach

6.2.1 Four Statistical Methods

We apply four different methods to identify dominant frames:

Method 1: Above Median

Procedure:

1. Calculate median of composite scores
2. Select frames above median

Rationale: Simple, robust, identifies top half

Method 2: Natural Break (Elbow Method)

Procedure:

1. Sort frames by composite score
2. Calculate differences between consecutive scores
3. Find maximum difference ("elbow")
4. Select frames above the break

Rationale: Identifies natural clustering in scores

Method 3: Cumulative Variance

Procedure:

1. Sort frames by composite score (descending)
2. Calculate cumulative sum of squared scores
3. Find point where 70% of total variance explained
4. Select frames up to that point

Rationale: Based on information theory, identifies frames carrying most information

Method 4: Statistical Outliers

Procedure:

1. Calculate mean and standard deviation of scores
2. Threshold = mean + 0.5 × standard deviation
3. Select frames above threshold

Rationale: Identifies statistically exceptional frames

6.2.2 The 3/4 Consensus Rule

A frame is declared dominant if selected by at least 3 of the 4 methods.

Example:

Frame	**Median**	**Elbow**	**Variance**	**Outlier**	**Dominant?**
Political	✓	✓	✓	✓	Yes (4/4)
Economic	✓	✓	✓	×	Yes (3/4)
Scientific	✓	×	✓	×	No (2/4)
Environmental	×	×	×	×	No (0/4)

6.3 Why This Approach Works

6.3.1 Theoretical Justification

1. **Multiple Dimensions**: Dominance is multifaceted
2. **Robustness**: No single method can create false positives
3. **Flexibility**: Adapts to different data distributions
4. **Transparency**: Clear, replicable criteria

6.3.2 Empirical Validation

We validated through:

- **Bootstrap analysis**: Results stable across resampling
- **Sensitivity analysis**: Robust to parameter variations
- **Cross-validation**: Consistent across time periods
- **Expert validation**: Aligns with qualitative assessments

7 Detailed Results and Interpretation

7.1 Overall Results (1978-2024)

Frame	**Info**	**Network**	**Causality**	**Prop.**	**Composite**	**Status**
Political	1.000	0.730	1.000	1.000	0.932	**Dominant**
Economic	0.587	0.713	0.464	0.722	0.622	**Dominant**
Scientific	0.705	0.348	0.841	0.465	0.590	Near-miss
Environmental	0.426	0.302	0.501	0.271	0.375	Not dominant
Justice	0.082	0.354	0.397	0.200	0.258	Not dominant
Cultural	0.099	0.348	0.206	0.232	0.221	Not dominant
Health	0.056	0.395	0.225	0.120	0.199	Not dominant
Security	0.000	0.500	0.000	0.000	0.125	Not dominant

7.2 Interpreting Each Frame's Profile

7.2.1 Political Frame: The Unequivocal Leader

Scores:

- Information: 1.000 (maximum information content)
- Network: 0.730 (strong hub position)
- Causality: 1.000 (drives all other frames)
- Proportional: 1.000 (highest sustained presence)

Interpretation: The political frame is the "conductor" of climate discourse. It:

- Provides maximum information about discourse structure
- Occupies central network position
- Causes other frames to appear
- Maintains highest, most consistent presence

7.2.2 Economic Frame: The Strong Second

Scores:

- Information: 0.587 (moderate information)
- Network: 0.713 (strong network position)
- Causality: 0.464 (moderate causal influence)
- Proportional: 0.722 (high sustained presence)

Interpretation: Economic frame achieves dominance through:

- Strong network centrality (co-occurs with political)
- High proportional presence
- Acts as "bridge" between political and other frames

7.2.3 Scientific Frame: The Declining Giant

Scores:

- Information: 0.705 (high information)
- Network: 0.348 (low network centrality)
- Causality: 0.841 (high causal influence)
- Proportional: 0.465 (moderate presence)

Interpretation: Scientific frame shows interesting pattern:

- High information and causality (still influences discourse)
- Low network centrality (marginalized in co-occurrence networks)
- Suggests transition from central to peripheral role
- "Near-miss" status reflects this ambiguous position

7.3 Temporal Evolution

7.3.1 Three Eras of Discourse

Era 1: Science-Led (1978-1989)

- Scientific frame dominant
- Mono-frame structure initially
- Climate as technical problem

Era 2: Transition (1990-1999)

- Economic frame emerges
- Triple-paradigm structure
- Competition for dominance

Era 3: Politics-Led (2000-2024)

- Political frame dominant
- Economic co-dominant
- Scientific marginalized to ¡10%

8 Conference Presentation Guide

8.1 Opening Strategy

8.1.1 The Hook

"We analyzed 266,000 articles over 47 years to answer one question: Which frames truly dominate climate discourse? Not just frequent frames, but those that structure how climate change is understood."

8.1.2 The Problem

"Previous studies counted frame frequency. But frequency $\neq$ dominance. A frame that appears 20% of the time might be more dominant than one appearing 30% if it carries more information, occupies central network positions, and drives other frames."

8.1.3 The Solution

"We developed a four-component framework examining frames through lenses of information theory, network analysis, causality, and proportional presence. Only frames excelling in 3+ dimensions are dominant."

8.2 Explaining Each Component

8.2.1 Information Theory (2 minutes)

Key Point: “Some frames are information hubs”

Analogy: “Like keywords in a conversation. When someone says ’budget’, you know economics is coming. When ’election’ appears, politics follows. Some frames predict entire discourse structure.”

Technical Detail: “We use mutual information and entropy reduction to quantify how much knowing one frame tells us about others.”

Result: “Political frame has maximum information content - knowing it’s present tells us the most about what else will appear.”

8.2.2 Network Analysis (2 minutes)

Key Point: “Frames form networks through co-occurrence”

Analogy: “Like social networks. Some people (frames) are influencers connected to everyone. Others are peripheral, connected to few.”

Technical Detail: “We use PageRank (Google’s algorithm) and eigenvector centrality to identify hub frames.”

Result: “Political and economic frames occupy central hub positions. Scientific frame increasingly peripheral.”

8.2.3 Causality Analysis (2 minutes)

Key Point: “Some frames cause others to appear”

Analogy: “Like dominoes. Push one (political), others fall (economic, environmental). We identify which frames are ’pushers’ vs ’pushed’.”

Technical Detail: “Granger causality tests if past values of X predict future Y. Transfer entropy captures non-linear information flow.”

Result: “Political frame shows highest causality - it drives appearance of other frames.”

8.2.4 Proportional Dominance (1 minute)

Key Point: “Sustained presence matters”

Analogy: “Not just who speaks most in meeting, but who consistently speaks more than average.”

Technical Detail: “Mean proportion, elevation above median, consistency, percentile rank.”

Result: “Political frame shows highest, most consistent presence.”

8.3 The Consensus Approach (1 minute)

Key Point: “Multiple methods ensure robustness”

Explanation: “Like having multiple judges. Frame must be selected by 3+ of 4 statistical methods to be dominant.”

Result: “Political and economic frames meet threshold. Scientific narrowly misses.”

8.4 Handling Difficult Questions

8.4.1 "Why equal weights?"

Answer: "Three reasons:

1. No theoretical justification for unequal weights
2. Sensitivity analysis shows results robust to reasonable variations
3. Transparency and replicability

We tested weights from 15-35% for each component. Results unchanged."

8.4.2 "Why 3/4 consensus?"

Answer: "Balance between stringency and flexibility:

- 4/4 too restrictive (only most extreme cases)
- 2/4 too lenient (methodological artifacts)
- 3/4 identifies robust dominance

This threshold empirically validated through bootstrap analysis."

8.4.3 "How does this relate to existing theory?"

Answer: "We operationalize Hall's paradigm theory. Hall described paradigms abstractly as 'systems of ideas.' We show paradigms are configurations of dominant frames, identified through our framework. This bridges theory-empirics gap."

8.4.4 "What about frames you don't show as dominant?"

Answer: "Non-dominant frames still important but play different roles:

- Conditional appearance (triggered by events)
- Supplement dominant frames
- Provide nuance without structuring discourse

Environmental frame, for example, appears during disasters but doesn't persistently structure discourse."

8.4.5 "Can this be applied to other topics?"

Answer: "Absolutely. Requirements:

- Longitudinal text data
- Frame annotations (manual or automated)
- Minimum 50 time periods for robust analysis

We're applying to immigration, healthcare, economic policy."

8.5 Closing Strategy

8.5.1 The Implications

"Our findings reveal troubling trend: scientific frame marginalized while political frame dominates. If media frames shape policy agendas, this suggests climate policy increasingly divorced from scientific evidence."

8.5.2 The Contribution

"We provide first systematic framework for measuring frame dominance. This moves beyond simple frequency counts to identify frames that truly structure discourse."

8.5.3 The Call to Action

"This framework is openly available. We encourage applications to other domains and welcome methodological improvements. Understanding how dominant frames shape public discourse is crucial for democratic deliberation."

9 Technical Implementation Guide

9.1 Data Requirements

Minimum Requirements

- **Temporal span:** Minimum 1 year (52 weeks)
- **Annotations:** Frame presence/absence or proportions
- **Granularity:** Weekly aggregation recommended
- **Frames:** 3-20 frames (8 optimal)

9.2 Software Stack

Python Implementation:

```
# Core libraries
import numpy as np
import pandas as pd
from scipy import stats
import networkx as nx

# Information theory
from sklearn.metrics import mutual_info_score
from pyinform import transfer_entropy

# Causality
from statsmodels.tsa.api import VAR
from statsmodels.stats.diagnostic import grangercausality
```

```python
# Network analysis
import networkx as nx

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

9.3 Computational Complexity

Component	Complexity	Bottleneck
Information Theory	$O(n^2)$	Pairwise MI calculation
Network Analysis	$O(n^3)$	Eigenvector centrality
Causality	$O(n^2 p)$	VAR model fitting
Proportional	$O(n)$	Simple statistics

Total runtime for 47 years, 8 frames: 5 minutes on standard laptop.

9.4 Validation Procedures

Validation Checklist

1. **Bootstrap confidence intervals**
   - Resample weeks with replacement
   - Compute dominance scores 1000 times
   - Calculate 95% confidence intervals
2. **Temporal stability**
   - Split data into periods
   - Check if dominant frames consistent
   - Identify transition points
3. **Sensitivity analysis**
   - Vary all parameters ±20%
   - Check if results change
   - Document sensitive parameters
4. **Qualitative validation**
   - Expert assessment
   - Close reading of high-scoring periods
   - Check face validity

10 Conclusion

This framework provides the first comprehensive approach to identifying dominant frames in discourse. By combining information theory, network analysis, causality testing, and proportional metrics, we move beyond simple frequency counts to identify frames that truly structure how issues are understood and debated.

The key innovation is recognizing that dominance is multidimensional. A frame might be frequent but not central, or central but not causal. Only frames that excel across multiple dimensions deserve the designation "dominant."

For researchers, this framework offers:

- **Theoretical grounding**: Operationalizes abstract concepts
- **Methodological rigor**: Multiple validation through different lenses
- **Practical applicability**: Clear implementation path
- **Reproducibility**: Detailed specifications for replication

For conference presentation, emphasize that this isn't just about measuring frames - it's about understanding how interpretive structures shape public discourse and, ultimately, policy responses to critical issues like climate change.